29/09/2020 MTH633 GROUP THEORY
FINAL, 2020, 5:00pm

MCQS'

1) $(Z_n, +)$ is a cyclic group with a generator ___ as
* 1

2) Let ϕ be a homomorphism function from group $(G, *)$ to group $(G', \cdot)$. Then, $(\ker\phi, *)$ is a ___ subgroup of $(G, *)$
* normal

3) The Group $G = Z/12Z$ is ___
* is not simple

4) The $Z_2 \times Z_2$ is ___
* cyclic * isomorphic

5) The center of a finite non-trivial p-subgroup G is ___
* non-trivial

6) Let P_1 and P_2 be sylow p-subgroups of a finite group G. Then P_1 and P_2 are ___ subgroups of G
* conjugate

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(Q7) Let G be a group of order p^n, $n \geq 0$ and Let X be a finite G-set. Then

* $|X| \equiv |X_G| \pmod p$

(Q8) $Z_3^* \geq 5$ _____ number of elements

* 15

(Q9) If _____ then factor group has no significant structure to supply about G

* $N = G$.

(Q10) Direction $\leftarrow$ or $\leftrightarrow$ gives us _____ function of (a inverse).

* inverse

(Q11) An automorphism of group G is ~~an~~ _____ of group G onto G.

* an isomorphism

(Q12) Let H and K be normal subgroups of a groups G with $K \leq H$, then G/K

* ~~G/H/H/K~~

* G/H.

(Q13) We define the join $H \vee N$ of H and N as the intersection of all subgroups of G that Contain HN, thus $H \vee N$ is smallest subgroup of G containing

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HN

* True

(Q14) Let X be a G-set. For $x_1, x_2 \in X$, Let $x_1 \sim x_2$ if and only if there exists $g \in G$ such that $gx_1 = x_2$. Then $\sim$ is an equivalence relation on X.

* equivalence.

(Q15) Let X be a G-set. For each $g \in G$, the function $\sigma_g : X \to X$ defined by $\sigma_g(x) = gx$ for $x \in X$ is a _____ of X

* permutation

(Q16) S_3 is not the group D_{30} of Symmetries of an equaliteral triangle.

* NO.

(Q17) Generally, for any sets A, B and C, we can view a map $*: A \times B \to C$ as defining a "multiplication," where any element a of A times any element b of B has value some element c of C.

* True.

(Q18) Which one is not rational

* $\sqrt{7}$ * $\frac{22}{7}$ * 0 * 5

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18) If N is normal subgroup of G, then G/N is abelian if and only if $\Leftrightarrow$ $C \leq N$.

* True

20) The inverse of $a^2b^4a^{-3}b^2a^5$ is

* $a^{-5}b^{-2}a^3b^{-4}a^{-2}$

21) $Z(G) = \{e\}$ in which case we say that the centre of G is trivial

22) Let H be a subgroup of a group G. Then H is normal if and only if for all a, b in G

* $(aH)(bH) = (ab)H$

23) Such subgroups K whose left and right cosets are very useful in studying a normal group

24) Every cyclic group of G is an abelian group.

25) D_4 is the group of Symmetries of the

* Square

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26) The nth dihedral group D_n is the group of symmetries of the regular _____

* n-gon

27) A _____ on S is a rule which assigns to any ordered pair (a,b) an element $a*b \in S$

* binary operation *

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